Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AGRO-HYPRO: Lightweight long-endurance hydrogen fuel cell agricultural UAV platform compliant with EU EASA regulations			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-generation Energy (), Bio-health ( ), Aerospace ( ), Quantum ( ), Advanced Robotics & Manufacturing (✓), AI (√), Cybersecurity (), Hydrogen (✓), Semiconductor materials ( )			
Industrial Technology Classification [1)]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Machinery & Materials		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Aerospace Systems	Fixed-wing/rotary-wing aircraft airframe	Renewable Energy	Fuel cells
Concept of Proposed Technology	Core concept: Co-develop and demonstrate a hydrogen fuel cell powered agricultural UAV (unmanned aerial vehicle) with substantially increased endurance and reduced airframe weight, while generating an evidence package for EU market entry (EASA UAS operations and hydrogen safety). [R1][R2][R3][R4] Key technical pillars requested in the AGRO-HYPRO RFP: [R1]			

	• Hybrid lightweight chassis combining CNC-machined aluminium (Al6061-T6) and carbon fibre (CFRP), with a joining concept targeting high rigidity and vibration resistance. [R1] • Low-pressure (<3 bar) solid hydrogen generation system based on an aluminium-water reaction, intended to reduce risk and mass compared with high-pressure storage (up to 700 bar). [R1] • High-efficiency composite propulsion: CFD-based optimisation of sandwich-structure CFRP propellers for aerodynamic efficiency. [R1] Relevant TNO capabilities (examples): • Powertrain and fuel cell verification at the Innovation Centre for Sustainable Powertrains in Helmond, including facilities for testing engines and fuel cells under controlled conditions. [R9][R19] • Battery and power electronics testing for hybrid architectures, including environmental testing capabilities. [R10] • Hydrogen technology and safety testing, including TNO's liquid hydrogen experimental test site for components such as valves, seals, composite tanks and pipelines. [R8] • Process safety expertise and testing infrastructure for assessing equipment and processes under extreme conditions, including explosions, temperature and pressure. [R18] • Vibration and structural dynamics testing via the TNO Structural Dynamics Lab (shakers, vibration measurements), relevant to validating airframe and propulsion vibration performance. [R12] • Lightweight materials and composite expertise, including thermoplastic composites and composite structural health monitoring research relevant to airframe design and inspection. [R11][R16] • Digital twin and autonomous drone experience from TNO projects combining drones and digital twin models for inspection applications. [R17] • Wind-tunnel test infrastructure available in the Netherlands via DNW for aerodynamic validation, which can be considered as part of the test plan. [R13] • Scientific expertise on fuel-cell powertrains, including work by TNO scientist Steven Wilkins and co-authors on fuel cell electric powertrains. [R14][R15]
Rationale for Joint R&D Project	The AGRO-HYPRO RFP identifies a trade barrier for Korean SMEs: despite having hydrogen and lightweight manufacturing technologies, EU entry is constrained by stringent aviation safety requirements and the lack of hydrogen safety verification data acceptable to EU stakeholders.

	[R1] Joint R&D with TNO addresses this by combining: • Korean SME design and manufacturing capabilities (airframe, composites, hydrogen and fuel-cell integration). [R1] • TNO verification and validation infrastructure and expertise (powertrain and hydrogen testing, safety engineering, modelling and simulation, and system verification planning). [R8][R9][R10][R12][R18][R19] • EU regulatory and safety-case development (EASA UAS regulatory gap analysis, SORA-based operational risk assessment, pre-compliance reporting and test evidence planning). [R2][R3][R4]
Key Words	1. Hydrogen fuel cell UAV 2. agricultural drone 3. Al-CFRP lightweight structure 4. CFRP propeller CFD 5. digital twin

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	1) Regulatory and certification readiness: perform an EU regulatory gap analysis for UAS operations (EASA Specific Category) and define a SORA-based safety-case plan and evidence needs. 2) System architecture and requirements: define UAV mission profile (agricultural payload, duty cycle), energy budget, and hydrogen-fuel-cell hybrid powertrain architecture. 3) Lightweight aero-structural design: co-design hybrid Al-CFRP chassis and composite propulsion components; create validated simulation models (CFD, structural dynamics). 4) Hydrogen safety engineering: hazard identification, quantitative risk assessment as appropriate, and design measures for low-pressure hydrogen generation and onboard integration; plan verification tests aligned with applicable standards and EU best practice. 5) Verification and validation plan: define test matrix, instrumentation, and data management needed for performance, durability, and safety demonstration in Korea and the Netherlands.			
R&D Target Specifications	Target performance (from AGRO-HYPRO RFP): [R1] • Endurance: 3 to 4 times longer than typical battery-based agricultural drones (RFP notes 20 to 30 minutes for battery drones). Derived endurance range: 60 to 120 minutes (20 to 30 minutes × 3 to 4), to be confirmed for the selected UAV class and payload. [R1] • Airframe mass: 30% reduction relative to a battery-drone baseline for the same mission class, via hybrid aluminium and CFRP structure. [R1] • Hydrogen system: low-pressure (<3 bar) hydrogen generation based on aluminium-water reaction; avoidance of high-pressure storage up to 700 bar. [R1] • Propulsion: sandwich-structure CFRP propellers optimised using CFD; target improved aerodynamic efficiency and reduced vibration loads. [R1][R12][R13] Regulatory deliverables: • Pre-compliance report and test evidence plan aligned with EU UAS rules (Regulation (EU) 2019/947) and EASA SORA guidance. [R2][R3][R4] • Hydrogen safety documentation aligned with EU ATEX framework (Directives 2014/34/EU and 1999/92/EC) and hydrogen safety engineering guidance. [R5][R6][R7]			

R&D Plan	Step 1: Pre-planning (6 months) (per AGRO-HYPRO RFP): Regulatory gap analysis and design optimisation. [R1] • EU regulatory consulting for UAS and hydrogen safety standards (ISO and ATEX) and definition of compliance strategy. [R1][R2][R3][R4][R5][R6][R7] • Digital twin and simulation concept for aero-structural design optimisation, including CFD and structural dynamics models. [R1][R17] • Definition of verification activities, facilities to be used (eg wind tunnel tests, hydrogen safety testing), and a phased demonstration plan. [R1][R8][R13][R18] Step 2: Main research (2.5 years) (per AGRO-HYPRO RFP): System integration and cross-verification. [R1] • Manufacture of a lightweight hydrogen UAV prototype, integration of hydrogen generation, fuel cell stack and BoP, hybrid energy storage, and avionics. [R1][R9][R10][R19] • Flight and field testing, including durability and versatility verification in Korean (rice farming) and Dutch (floriculture and field farming) environments. [R1] • Iterative design updates supported by test data, and completion of the safety-case and compliance evidence package for EU market entry. [R2][R3][R4]

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) responsibilities: • Lead EU regulatory gap analysis for UAS (EASA Specific Category) and structure the SORA-based operational risk assessment approach. [R2][R3][R4] • Develop and validate digital models (CFD, structural dynamics, thermal and energy management) and support design optimisation. [R12][R13][R17][R19] • Provide hydrogen safety engineering expertise, including hazard identification, safety requirements, and test protocols. [R7][R18] • Execute or support verification testing via TNO facilities (powertrain and hydrogen testing, vibration testing, safety testing) and through access to external test infrastructure when needed (eg wind tunnel). [R8][R9][R10][R12][R13][R18][R19] • Consolidate test and simulation results into a pre-compliance report and evidence package structure. [R2][R3][R4]
Expected R&R of Korean SME Partner	Korean SME Partner responsibilities: • Provide baseline UAV design, mission requirements, and manufacturing approach for hybrid Al-CFRP airframe and composite components. • Provide hydrogen fuel cell stack and balance-of-plant (BoP) know-how and integrate the low-pressure hydrogen generation concept. • Manufacture prototypes and support integration, instrumentation, and iterative design improvements. • Lead agricultural application demonstrations in Korea and support cross-validation trials in the Netherlands. • Contribute to technical documentation and data packages needed for verification, certification readiness and commercialisation.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical outcomes: • Validated system architecture and digital twin models for a hydrogen fuel cell agricultural UAV, including aero-structural and energy management models. [R17][R19] • Prototype-level verification plan and test evidence aligned to EU UAS and hydrogen safety expectations, reducing technical barriers to trade. [R2][R3][R4][R5][R6][R7] • Demonstrated endurance increase (target 3 to 4 times vs battery drones) and weight reduction (target 30%) as specified in the AGRO-HYPRO RFP, supported by test data. [R1] Impact: • Accelerated EU market entry readiness for the Korean SME Partner through structured compliance evidence (SORA-based safety-case planning and hydrogen safety documentation). [R2][R3][R4][R7] • Strengthened EU Korea collaboration on sustainable agriculture and zero-emission aerial mobility. [R1] • Creation of a reusable verification and certification pathway and reference models for hydrogen-powered UAV platforms. [R2][R3][R4]

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization